

Co2 AT, M.

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 100

Annexure A

Code of Conduct:

I Hemanth P(192524297) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



① RAS Encryption Analysis.

Given:

$$P = 11, q = 13, e = 7, \text{ Message } M = 9$$

a) Compute n & $\phi(n)$:

$$n = P \times q = 11 \times 13 = 143$$

$$\phi(n) = (P-1)(q-1) = 10 \times 12 = 120$$

$$n = 143$$

$$\phi(n) = 120$$

b) find the private key d ,

$$Td = 1 \pmod{120}$$

$$T \times 103 = 721$$

$$721 \div 120 = 6 \text{ remainder } 1$$

$$d = 103$$

$$\text{private key} = (103, 143).$$

c) Encrypt the message.

$$C = M^e \pmod{n}$$

$$C = 9^7 \pmod{143}$$

$$9^2 = 81$$

$$9^4 = 81^2 = 6561$$

$$9^7 = 9^4 \times 9^2 \times 9$$

$$= 481 \pmod{143}$$

$$C = 48$$

d) decrypt:

$$M = C^d \bmod 143$$

$$M = 48^{103} \bmod 143$$

$$M = 9$$

e) effect of changing e

* Small value of e (3 or 17) makes encryption faster.

* $e = 65537$ is widely used.

2. Block cipher Mod evaluation.

$$P = [1010, 1010, 1111, 1010]$$

$$\text{key} = 1100$$

$$\text{iv} = 0000$$

encryption is performed using XOR

ECB mode

$$C = P \oplus K$$

block 1,

$$(1010) \text{ XOR } (1100) = 0110$$

block 2,

$$(1010) \text{ XOR } (1100) = 0110$$

block 3,

$$(1111) \text{ XOR } (1100) = 0011$$

CBC mode,

block 1,

$$1010 \oplus 0000 = 1010$$

$$1010 \oplus 1100 = 0110$$

$$C1 = 0110$$

block 2

$$1010 \oplus 0110 = 1100$$

$$1100 \oplus 1100 = 0000$$

$$C2 = 0000$$

block 3,

$$1111 \oplus 0000 = 1111$$

$$1111 \oplus 1100 = 0011$$

$$C3 = 0011$$

block 4,

$$1010 \oplus 0011 = 1001$$

$$1001 \oplus 1100 = 0101$$

$$C4 = 0101$$

CBC cipher text

[0110, 0000, 0011, 0101]

3). DES vs 3-DES vs blowfish

gm:

Algorithm	key size	Time/Block
DES	56 bits	2ms
3-DES	168 bits	6ms
blowfish	128 bits	1ms

for 1000 blocks

DES

$$1000 \times 2 = 2000 \text{ ms}$$

$$= 2 \text{ s}$$

3-DES

$$1000 \times 6 = 6000 \text{ ms}$$

$$= 6 \text{ s}$$

blowfish

$$1000 \times 1 = 1000 \text{ ms}$$

$$= 1 \text{ s}$$

Algorithm	Total Time	Security
DES	2s	Low
3-DES	6s	High
blowfish	1s	High

④ Given: Prime $P=23$, Generator $g=5$, user A private key $=6$, user B private key $=15$

Public key of A

$$A = 5^6 \bmod 23$$

$$5^6 = 15625$$

$$15625 \bmod 23 = 8.$$

$$A = 8$$

Public key of B

$$B = 5^{15} \bmod 23$$

$$5^{15} \bmod 23 = 19$$

$$B = 19$$

$A \leftrightarrow A \text{ Hacker}$

$A \text{ Hacker} \leftrightarrow B$

Prevention:

- * digital signature
- * certificates
- * Public Key Infrastructure (PKI)

5. given:

RSA key size = 1024 bits

ECC key size = 160 bits

Computational cost

Parameters	RSA	ECC
Key size	1024 bits	160 bits
Encryption speed	Moderate	Fast
Decryption speed	Slow	Fast
Memory usage	High	Low
Power consumption	High	Low

Power and Memory:

• ECC usage much smaller key requiring:

- * Less RAM
- * Less Storage
- * Lower CPU usage
- * Lower battery power

Justification:

- * provides equivalent security
- * with smaller key
- * consumes less power.